Ranking Transfers, Thresholds Don't: Unsupervised Test-Time Adaptation for Cross-Corpus Audio Deepfake Detection

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***Abstract*—Audio deepfake detectors generalise poorly across corpora despite near-perfect in-domain accuracy. We show that a fine-tuned self-supervised detector's *ranking* (ROC-AUC) largely transfers cross-corpus even when its decision *threshold* does not calibrate, and that this precondition, not the target corpus itself, predicts when adaptation helps: EER gain correlates with source-domain AUC at $r{=}{+}0.47$ ($p{=}0.002$) across 40 (target, seed) runs. Where the precondition holds, the model's most-confident predictions on unlabeled target audio are reliable enough to self-train on, which we exploit with a stable unsupervised *test-time adaptation* (TTA) method: self-train on confident-ranked pseudo-labels and enforce channel-consistency, adapting only LayerNorm and the classifier head, with no target labels and no source data at test time. On a leave-one-corpus-out protocol over four targets (ASVspoof2019, a LibriSpeech-based TTS corpus, In-the-Wild, and a multilingual Arabic set), ten seeds each, this delivers EER gains where source AUC is ≥ 0.84 (In-the-Wild $12.78 \to 11.33\%$, Arabic $22.50 \to 20.99\%$, ASVspoof2019 $5.03 \to 3.47\%$, each significant by a *per-target* Wilcoxon test, $p{\leq}0.03$) and is EER-neutral on the one target below that threshold (LibriSpeech-TTS, source AUC 0.73). Entropy-minimisation TTA (Tent) is unpredictably catastrophic—four of ten ASVspoof2019 seeds at or near chance, with no advance warning and a standard deviation as large as its mean; a non-SSL backbone collapses on all four targets. A train-time domain-adversarial baseline (DANN) and an aggregation-separation baseline (ASDG-style) are both beaten pooled over all 40 (target, seed) pairs ($p{<}10^{-4}$ each) despite privileged access to target data during training, though per target the margin is confined to Arabic and ASVspoof2019. A proxy $\mathcal{A}$-distance analysis shows domain separability does *not* decrease after adaptation, ruling out domain-invariance as the mechanism and supporting our ranking-calibration account instead.**

***Index Terms*—audio deepfake detection, anti-spoofing, test-time adaptation, cross-corpus generalisation, self-supervised learning, EER.**

I. Introduction

Self-supervised (SSL) front-ends such as wav2vec 2.0 and XLS-R, paired with lightweight back-ends, reach very low in-domain error on standard anti-spoofing benchmarks [1]–[3]. Deployed on audio from a different source—new microphones, codecs, languages, or generators—accuracy drops steeply, a well-documented generalisation gap [5], [6], [11].

We start from a diagnostic measurement: taking a fine-tuned XLS-R detector and evaluating it on an unseen corpus, its **ROC-AUC stays high while its accuracy at a fixed threshold collapses**—the model still *ranks* clips correctly but its operating point is mis-calibrated for the new domain. Because ranking transfers, the model's most-confident predictions on unlabeled target audio are trustworthy, so we can use them to adapt the model to the target domain *at test time*, without any labels.

Our contributions: (i) the empirical finding that ranking transfers cross-corpus even when the threshold does not, and its consequence for unsupervised adaptation; (ii) a simple, stable TTA method—confident-ranked pseudo-label self-training plus channel-consistency, adapting only LayerNorm and the head; (iii) a four-target, multilingual, ten-seed leave-one-corpus-out evaluation against seven comparison points (source-only, Tent, BatchNorm-only recalibration, self-training-only, a non-SSL backbone, DANN, and an ASDG-style aggregation-separation baseline), including the one target where the method does not help; (iv) a mechanistic analysis—proxy $\mathcal{A}$-distance before/after adaptation—showing the gain does *not* come from erasing domain identity.

Related work. Frozen or fine-tuned SSL front-ends with AASIST or attentive back-ends dominate recent benchmarks [1]–[4]; we additionally test whether the cross-corpus gap is an SSL question at all via a from-scratch, non-SSL backbone. Train-time domain generalisation (DG) methods align real and separate fake across source domains (aggregation-separation [5]) or use gradient-reversal domain-adversarial training (DANN) [8]; these need target data *during training* rather than at deployment, and we show a much cheaper test-time alternative is not obviously worse. Entropy minimisation (Tent) [7] is the standard TTA recipe in vision but, as we show, is unstable here; batch-statistics recalibration [9] is a no-op for LayerNorm architectures. We borrow the proxy $\mathcal{A}$-distance from Ben-David et al. [10] to test mechanism, and use ASVspoof2019-LA, In-the-Wild [11], and multilingual MLAAD as evaluation corpora.

II. Method

Let x be a waveform with label $y \in \{0,1\}$ (fake=1). A detector f_θ pools frame features into an embedding e and outputs $s_\theta(x) = \sigma(w^\top e + b) = P_\theta(y{=}1 \mid x)$. The encoder is XLS-R-300M with only its top four transformer layers trainable, followed by attentive temporal pooling and a linear head. Its normalisation layers are *LayerNorm*, which

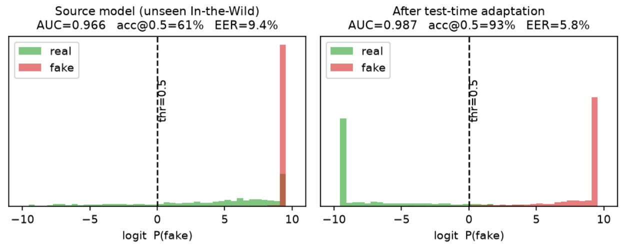

Fig. 1. Detector scores on unseen In-the-Wild audio. **Left:** the source model pushes many real clips past τ=0.5 (accuracy well below AUC) yet still ranks most real below fake. **Right:** TTA shifts the real mass back across the threshold.

carries no running batch statistics—unlike BatchNorm—so a batch-statistics-recalibration baseline is architecturally a no-op here, and confidence-driven gradient adaptation is the minimal mechanism that can move these layers at all.

The ranking–calibration gap. Ranking quality $\mathrm{AUC} = \Pr[s_\theta(x^+) > s_\theta(x^-)]$ is threshold-free and invariant to reparametrisation of s_θ; accuracy at a fixed threshold τ is not. Empirically, AUC degrades far less than accuracy at τ=0.5 under source→target shift (Fig. 1). We therefore trust score *order*: the extreme quantiles of unlabeled target scores are higher-precision pseudo-labels than a fixed threshold, even when s_θ is mis-calibrated—though this premise weakens when source AUC on the target is itself low (Sec. IV). We scope the method accordingly: it targets *moderately class-balanced, high-initial-AUC* adaptation regimes—the fixed-quantile split Q_q vs. Q_{1-q} implicitly assumes both a reliable source ranking and roughly balanced target classes, and Sec. IV and Sec. V show what happens when each assumption breaks.

Test-time adaptation. Each epoch, we assign confident pseudo-labels from the q- and $(1-q)$-quantiles of target scores, ignoring the middle; self-train with cross-entropy on this confident set ($\mathcal{L}_{\mathrm{ST}}$); and enforce prediction invariance under a stochastic channel perturbation $a(\cdot)$ (gain+noise) via an MSE consistency term $\mathcal{L}_{\mathrm{cons}}$ between clean and perturbed views (stop-gradient on the clean target). The objective is

$$\min_{\{\gamma,\beta\},\,\phi} \; \mathcal{L}_{\mathrm{ST}} + \lambda\,\mathcal{L}_{\mathrm{cons}}, \qquad \lambda{=}0.3,\; q{=}0.3,\; E{=}4, \qquad (1)$$

optimising *only* the LayerNorm affine parameters $\{\gamma, \beta\}$ of the fine-tuned layers and the head ϕ; the rest stays frozen. Restricting the trainable set and using only confident tails prevents the degenerate collapse to which entropy minimisation is prone.

Domain divergence. To test whether adaptation works by making source and target indistinguishable in embedding space (as adversarial DG methods explicitly optimise for), we compute a proxy $\mathcal{A}$-distance $\hat{d}_{\mathcal{A}} = 2(1-2\epsilon_h)$ following Ben-David et al. [10], where ϵ_h is a linear domain classifier's held-out error on frozen embeddings, before and after adaptation.

III. Experimental Setup

Protocol. Leave-one-corpus-out over four EER-capable targets: ASVspoof2019-LA, a LibriSpeech-based multi-system-TTS corpus (D2), In-the-Wild [11], and Arabic deepfakes (ArAD). For each held-out target, the source pool merges the other three corpora with a 38-language MLAAD subset (fake-only) for generator/language diversity; held-out target audio, including its MLAAD-overlapping language content, is never in any source pool, and six MLAAD languages are held out of every source pool entirely to test genuine unseen-language generalisation.

Comparison points. Source-only (no adaptation); Tent [7] (entropy minimisation, same trainable set as ours); BN-only (AdaBN-style recalibration [9], no gradient step); self-training-only (ours with λ=0); Oracle (fine-tuned on target labels, a supervised upper bound); RawNet2Lite (a from-scratch, non-SSL backbone in the RawNet2 family [4], trained identically, to isolate whether robustness comes from the SSL front-end); DANN [8] and an ASDG-style aggregation-separation baseline [5] (both trained to convergence with the held-out target's *unlabeled* inputs available during source training—strictly more privileged access than our method has).

Implementation. 16 kHz audio, top-4-layer XLS-R fine-tuning, Adam lr 2×10^{-4}, batch 32, source trained 8 epochs. All results are mean $\pm$ std over ten seeds on every target, for our method and for every baseline it is compared against.

IV. Results

Table II is the main result. Our method wins on three of four targets: ASVspoof2019 ($5.03 \to 3.47\%$, 10/10 seeds), Arabic ($22.50 \to 20.99\%$, 10/10), In-the-Wild ($12.78 \to 11.33\%$, 8/10); on LibriSpeech-TTS it is EER-neutral ($33.54 \to 33.74\%$, improved in only 5/10 seeds), the target with the lowest source AUC (0.73 vs. ≥ 0.84 elsewhere)—consistent with our ranking-precondition account: across all 40 (target, seed) points, source AUC and EER gain correlate at r=+0.47 (p=0.002, Fig. 2).

Significance. Ten seeds per target make a *per-target* Wilcoxon signed-rank test possible: at n=5 the combinatorial floor is 0.0625, so no per-target claim was attainable at any effect size, whereas at n=10 it is 0.002. Against source-only, ours is significant on ASVspoof2019 (p=0.0020), Arabic (p=0.0020) and In-the-Wild (p=0.0293), and not on LibriSpeech-TTS (p=0.23), which is the EER-neutral target. Pooled over all 40 (target, seed) pairs the method beats source-only at p=2.4×10^{-5}, Tent at p=1.5×10^{-7}, self-training-only at p=0.0056, and both DG baselines at p<10^{-4} (30/40 pairs each). Per-target 95% bootstrap CIs (10k resamples over eval clips) on mean EER are non-overlapping between source and ours on ASVspoof2019 ($[4.27, 5.82]$ vs. $[2.97, 3.99]$), In-the-Wild ($[12.06, 13.50]$ vs. $[10.53, 12.21]$) and Arabic ($[21.36, 23.53]$ vs. $[20.12, 21.79]$); they overlap on LibriSpeech-TTS, consistent with the EER-neutral finding above.

Tent is not merely unstable, it is unpredictably catastrophic. Its ASVspoof2019 per-seed EER is 4.16, 3.91, 3.04, 49.52, 3.42, 49.44, 49.69, 42.11, 25.31, 4.12%—five seeds close to source-only, four at or near chance and one midway, with no warning sign; the standard deviation (21.96) is comparable to the mean (23.47). Doubling the seed count did

TABLE I

TEST-TIME ADAPTATION VS. TWO TRAIN-TIME DOMAIN-GENERALISATION BASELINES, MEAN EER % / AUC OVER THE SAME 10 SEEDS FOR EVERY METHOD. BEST PER ROW IN **BOLD**. OURS IS BETTER POOLED OVER ALL 40 (TARGET, SEED) PAIRS AGAINST BOTH (30/40, $p<10^{-4}$ EACH), BUT PER TARGET THE ADVANTAGE IS CONFINED TO ARABIC AND ASVSPOOF2019; ASDG IS BETTER ON LIBRISPEECH-TTS. ASDG RAN LOCALLY ON DIFFERENT HARDWARE, SO ITS WALL-CLOCK TIME IS OMITTED RATHER THAN COMPARED DIRECTLY.

Target	Source	DANN [8]	ASDG [5]	Ours
ASVspoof2019	5.03 / .989	7.35 / .966	4.98 / .987	**3.47** / .994
LibriSpeech-TTS	33.54 / .727	34.43 / .717	**32.64** / .749	33.74 / .727
In-the-Wild	12.78 / .945	13.34 / .924	17.33 / .891	**11.33** / .958
Arabic	22.50 / .843	28.53 / .759	27.41 / .798	**20.99** / .866

not resolve the behaviour into a stable average, it revealed the collapse to be routine rather than a single unlucky run. On Arabic it is a *consistent* failure instead ($47.35\pm5.03\%$, chance-level in $9/10$ seeds). Our method shows no comparable collapse on any target (largest std 2.01, on the one target it does not help).

Backbone and prior DG baselines. RawNet2Lite (no SSL pretraining) stays near chance on every target (AUC 0.40–0.65), confirming cross-corpus viability starts from SSL pretraining, not the TTA step; retraining it for 40 epochs with stronger regularisation does not close the gap, ruling out undertraining as the explanation. Both train-time baselines are beaten pooled over all 40 (target, seed) pairs ($30/40$, $p<10^{-4}$ each) despite privileged target access, but at ten seeds the per-target picture is narrower than a three-seed comparison suggested: our margin is significant on Arabic and ASVspoof2019 against both, while on In-the-Wild we tie DANN ($5/10$ seeds) and on LibriSpeech-TTS ASDG is the better method (32.64% vs. our 33.74%). DANN is markedly worse on Arabic (28.53% vs. our 20.99%), $3\times$ more expensive (~30 vs. ~9 min per target-seed), and the least seed-stable method we test (ASVspoof2019 EER ranges 3.23–14.85% across seeds). ASDG beats DANN on three of four targets but beats plain source-only fine-tuning on only two of four (Table I), with a monotonic seed-to-seed drift on In-the-Wild ($13.90, 19.78, 28.05\%$) reminiscent of Tent's instability. Neither train-time baseline is a reliable route to cross-corpus robustness for this task, and a lighter-weight test-time mechanism is not just cheaper but also more effective.

Mechanism. An inductive check (adapt on one half of the target, evaluate on the disjoint half) confirms the gain is not transductive memorisation. On *every* target, the proxy $\mathcal{A}$-distance and a linear domain-origin probe both increase slightly after adaptation rather than decreasing (e.g. Arabic $\hat{d}_{\mathcal{A}}: 1.545 \to 1.606$)—the opposite of what domain-invariance would predict, even on targets where EER improves substantially. Adaptation re-shapes the decision boundary using rank-reliable pseudo-labels; it does not erase domain identity, which is independently consistent with why DANN underperforms despite explicitly optimising that quantity. A 38-language MLAAD breakdown shows fake-clip recall is uniform across seen and unseen languages (93.9% vs. 93.3%), so multilingual

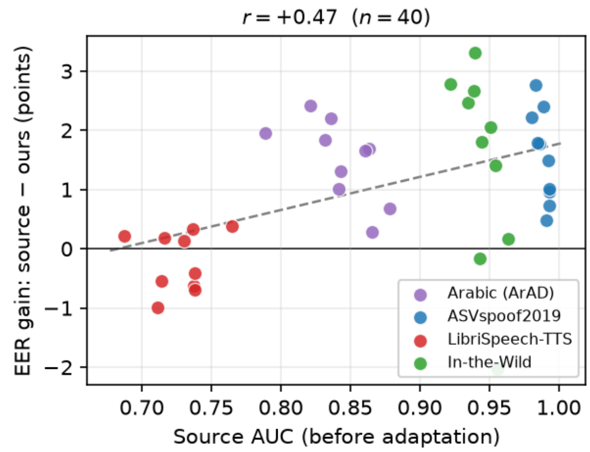

Fig. 2. EER gain (source−ours) vs. source AUC before adaptation, one point per (target, seed), all 40 runs. LibriSpeech-TTS (red), the lowest-AUC target, clusters at or below zero gain; higher-AUC targets cluster above it ($r=+0.47$, $p=0.002$).

TABLE II

CROSS-CORPUS EER % / AUC, MEAN OVER 10 SEEDS (LEAVE-ONE-CORPUS-OUT, TRANSDUCTIVE). BEST OF {SOURCE, TENT, ST-ONLY, OURS} PER COLUMN IN **BOLD**.

Method	ASVspoof2019	LibriSpeech-TTS	In-the-Wild	Arabic
Source-only	5.03 / .989	33.54 / .727	12.78 / .945	22.50 / .843
Tent [7]	23.47±21.96 / .776	35.44±3.20 / .674	27.34±11.21 / .735	47.35±5.03 / .527
Self-training only	4.20 / .987	**33.39** / .709	11.55 / .950	22.19 / .846
Ours (TTA)	**3.47** / .994	33.74 / .727	**11.33** / .958	**20.99** / .866
Oracle (target-supervised)	0.89 / .999	13.93 / .939	4.42 / .992	9.50 / .968

robustness is not an artefact of language coverage during training.

V. DISCUSSION AND LIMITATIONS

The method is transductive (unlabeled target inputs at adaptation time, the standard TTA setting); an inductive check tracks the transductive numbers closely on every target, including reproducing the LibriSpeech-TTS null result. We report per-target, per-seed numbers throughout rather than a single aggregate, so that the limitations below can be read against the evidence:

- **Needs an already-reliable source ranking.** A from-scratch or low-AUC starting point is not a regime we can currently rescue (Fig. 2).
- **Class balance is a second precondition, alongside AUC.** On the regime matching the official ASVspoof2021-DF protocol (train on 2019-LA alone), source-only reaches 26.33% EER and TTA actively *harms* (42.46%) under that evaluation set's 97.5%-spoof imbalance. The cause is structural, not incidental: the fixed-quantile rule labels the bottom q of the pool "confident real" whatever the true prior, so under severe skew that bucket fills with mislabelled spoof and self-training teaches the model that much of the fake class is real. Two label-free corrections fail to recover it—a prevalence-weighted budget helps under one source

recipe (32.58%) but not another (42.46%), and adding a curriculum schedule over q changes nothing (42.50%), in both cases still worse than not adapting at all. We therefore state moderate class balance as a precondition of the method rather than an implementation detail.

- **Not a state-of-the-art detector.** Our source model is an XLS-R encoder with a light attentive head trained on a leave-one-corpus-out pool, not a system tuned for any single benchmark, and published anti-spoofing results on the standard ASVspoof protocols reach substantially lower absolute EER than the source-only column here. The claim of this paper is the *relative* gain from test-time adaptation on a fixed backbone, measured against that same backbone unadapted under an identical protocol; it should not be read as a competitive absolute result on any individual benchmark, and the mechanism would have to be re-verified on a stronger source model before it could be.
- **Fixed hyperparameters.** q, λ, E are chosen without target labels and are not per-target optimal. A label-free tail-separation statistic predicts Δ_2 strongly but with the sign inverted from a reliability reading (r=$+0.74$, p=0.006): saturation, not unreliability, is what makes the consistency term freeze an overconfident boundary—a calibration effect, consistent with the ranking-calibration account. Acting on it does not pay at our scale (fixed λ is within 0.24 EER of a per-point oracle), so λ stays constant.
- **Scale.** Four targets, ten seeds. Growing from three to ten seeds did not uniformly strengthen our claims: it made the per-target significance tests possible at all, and confirmed Tent's collapse as routine rather than a single unlucky run, but it also shrank the margin over the train-time DG baselines from a three-of-four sweep to a pooled result with two clear per-target wins. Four corpora remain few enough that the $\mathrm{AUC} \geq 0.84$ precondition is calibrated on a handful of points.
- **SSL dependency.** The gain requires an SSL-pretrained backbone (Sec. IV); it is an adaptation layer on top of an already-transferable representation, not a backbone-agnostic recipe.

VI. Conclusion

Cross-corpus audio deepfake detection fails not because ranking is lost but because the operating point is mis-calibrated, and whether that ranking survives (source AUC $\gtrsim 0.84$) is itself the predictor of whether adaptation will help—a precondition we state as a contribution, not a caveat. A simple unsupervised test-time adaptation exploiting this improves generalisation, significantly by per-target Wilcoxon test, on the three of four leave-one-corpus-out targets meeting that precondition, and is neutral on the fourth. Entropy-minimisation TTA is unpredictably catastrophic, a non-SSL backbone collapses outright, and two independent train-time domain-generalisation baselines underperform ours despite privileged data access and $3\times$ the compute. A proxy divergence analysis shows the mechanism is re-calibration of an already class-organised representation, not domain-invariance.

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